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Process study of picosecond laser welding of soda-lime glass

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Highlights

- Appropriate laser power is the basis for forming a reliable connection. Low power will lead to insufficient energy to form a weld; high power will cause an increase in defects.
- The scanning spacing and scanning times are too large or too small to have a negative effect on the strength of the weld.
- The maximum breaking strength is obtained when the scanning direction is perpendicular to the force direction of the test.
- Effective welds can be formed with focal points located up to 200 μm above and below the interface, with the greatest strength obtained at the interface.

Declaration of competing interest

Data availability

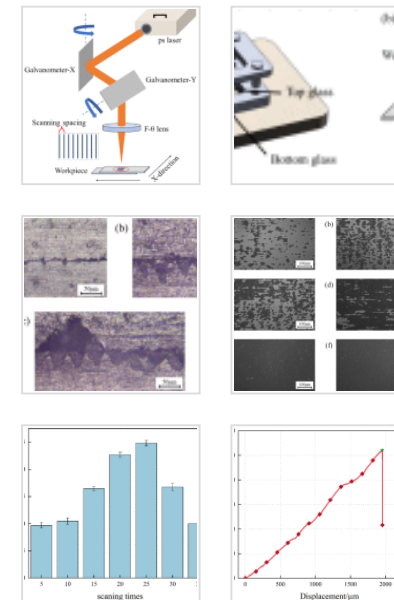
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Abstract

The various excellent properties of glass make it widely used in aerospace, solar cells and other fields, and its processing methods have also attracted much attention. Ultra-fast laser has the characteristics of high peak energy density and low heat impact, which has unique advantages in welding of transparent materials such as glass. At this stage, most of the ultra-fast laser welding is carried out for the glass with optical contact, and the harsh optical contact conditions hinder the further application of glass welding technology. In this paper, an attempt is made to weld non-optical contacted soda-lime glass using high repetition frequency infrared picosecond laser. The effects of laser power, scanning times, scanning spacing, scanning direction and focus position on weld morphology and breaking strength were systematically investigated. The results show that the scanning times and scanning spacing affect the volume of glass melt on one hand, and the stress magnitude during the welding process on the other hand, which have a greater impact on the overall weld strength. Under the optimal process parameters, an effective connection can be achieved within $\pm 200\mu\text{m}$ of the defocusing amount, and the weld formed has a size of 10mm $\times$ 5mm. A strength of 24.85MPa is obtained when the force direction is perpendicular to the scanning direction. The mechanism of glass welding was finally

Figures (13)



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Tables (1)

 Table 1. Parameter type and range.

revealed, providing technical guidance for the practical application of ultrafast laser welding of glass.

Keywords

Picosecond laser; Laser processing; Glass welding

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1. Introduction

Glass is a typical hard and brittle material characterized not only by high hardness, low coefficient of thermal expansion and good chemical stability, but also by excellent biocompatibility, and is widely used in aerospace, solar cells, microelectronics, energy, and biomedical applications [1], [2], [3]. In practical industrial applications, glass materials often require multiple assemblies or connections before they can be used, however, high hardness and brittleness make them difficult to process [4], [5], [6]. Conventional methods of joining glass are mainly adhesive bonding [7], anodic bonding [8], solid bonding and fusion welding [9]. Each of these methods has its own drawbacks, for example, the life of the adhesive is easily affected by the surrounding environment; anodic bonding requires materials with similar coefficients of thermal expansion; solid bonding takes longer and requires higher surface quality; fusion welding is characterized by a high-temperature phase transition, which makes the weld material susceptible to rupture, and so on [10], [11].

Ultrashort pulsed lasers have demonstrated unique advantages in many areas [12], [13], [14], [15], [16], [17], including welding of transparent materials [18], [19], due to their small heat-affected zones and high processing accuracy. When the intensity of the ultrashort pulse laser exceeds the threshold value of transparent materials, it will

generate nonlinear heads inside the material, causing the local material near the focal point to melt and thus realize the connection [20], [21], [22].

In 2005, T. Tamaki et al. realized the welding of quartz glass for the first time using a femtosecond laser, and since then, the welding of transparent materials such as glass has been widely investigated [23]. In 2006, W. Watanabe et al. used a femtosecond laser for the welding of quartz glass and borosilicate glass, and achieved high-quality welding of dissimilar glasses by varying the translational speed and the laser pulse energy, with a maximum joint strength of 15.3MPa [24]. In the years since, researchers have carried out a great deal of work on glass welding, but most of these studies were conducted under experimental conditions with optical contact. In 2015, J. Chen et al. achieved welding of fused silica without optical contact using a picosecond laser at 1030nm. It was found that the maximum weldable gap in the welding of glass without optical contact was not significantly affected by the welding speed, but strongly depended on the laser power and the focus position. And five different types of material modifications (plasma ablation, heat-affected zone ablation, intermittent welding, continuous welding, and heat-affected zone under the interface) were observed in a series of experiments [25]. In 2017, H. Tan et al. performed the welding of fused quartz glass using a 1030nm femtosecond laser at 600kHz and investigated the effect of different focus positions on the strength of the weld. When the laser pulses were focused below the interface for optical-contact, the maximum bond strength of 58.2MPa was obtained [26].

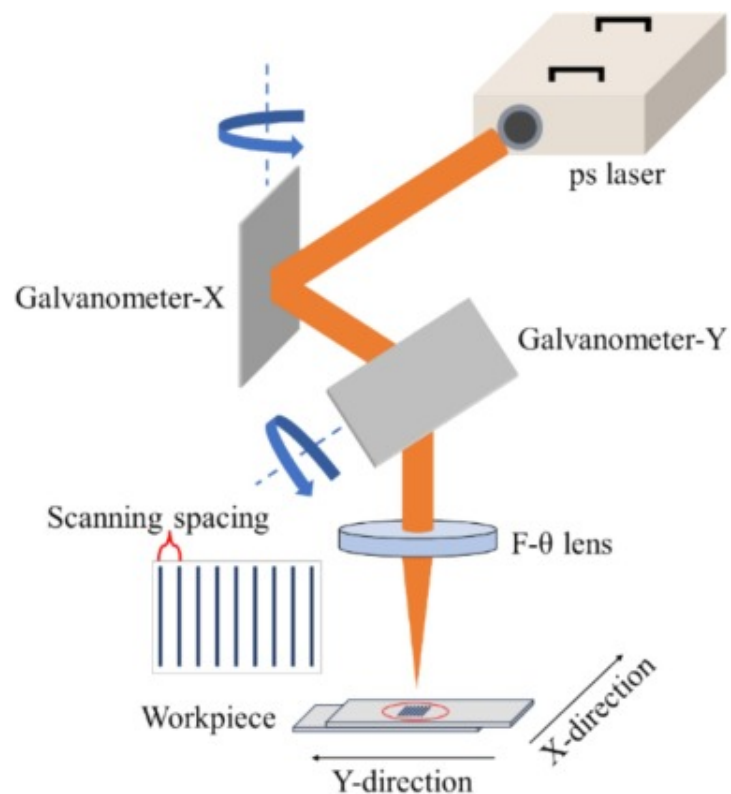
In fact, most of the existing studies on glass welding have focused on the laser power and the amount of defocusing, neglecting the effect of factors such as the scanning times and scanning spacing on the weld. Multiple scans seem to be more favorable to increase the material melt for a more stable bond, but may also increase the interfacial stress. The scanning spacing may also have a greater effect on the processing efficiency and the overall strength of the weld.

Therefore, in this paper, a 1064nm picosecond laser was used to achieve the welding of non-optical contact transparent glass. The effects of laser power, scanning times, scanning

spacing, scanning direction and focusing position on weld morphology and fracture strength were mainly investigated. Two glass samples with non-optical contact were effectively connected by several scans in a 10mm×5mm area. The mechanism of glass welding was finally revealed, which provides guidance for further applications of ultrafast laser welding of glass.

2. Experimental procedure

The processing system mainly consists of a laser, a 2D galvanometer and a F- θ lens, as shown in [Fig. 1](#). The Edgewave PX200-2GFH laser was used for the welding test, with a wavelength of 1064nm, a pulse width of 12ps, a maximum repetition frequency of 50MHz, and a maximum output power of 120W. Combined with the focusing lens and 2D galvanometer, the laser welding was carried out, and the maximum machining range of a single process was 100mm×100mm. the movement of the laser focal plane was realized by adjusting the Z-axis, and the Z-axis movement accuracy of approx. 1 μ m. The main process parameters and their ranges are shown in [Table. 1](#). The scanning direction is unidirectional along the X-axis unless otherwise specified.



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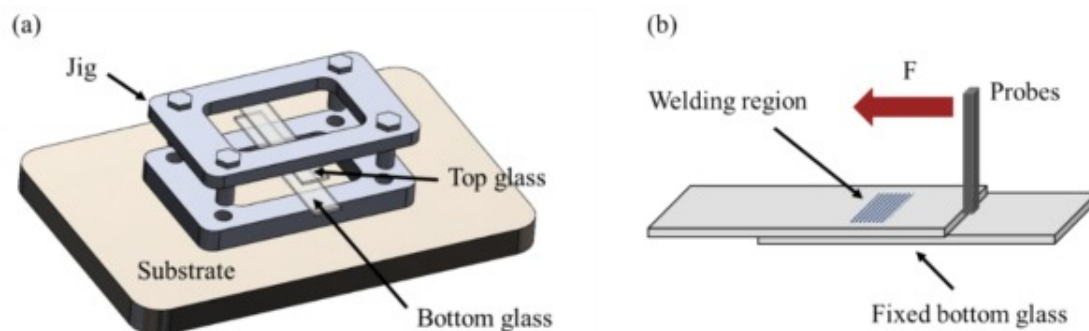
Fig. 1. The schematic diagram of the experimental system.

Table 1. Parameter type and range.

Type	Range
Frequency/kHz	50
Power/W	1.5,2.5,3.5
Scanning speed/mm/s	100
Scanning times	5~50

Type	Range
Scanning spacing/ μm	5~30
Scanning direction	unidirectional scanning, cross scanning
Defocus/mm	-0.2~+0.2

The glass used in the experiments was soda-lime glass with dimensions of $75 \times 25 \times 1 \text{ mm}^3$. The glass was ultrasonically cleaned in an alcohol solution for 5 min before processing, and the surface was wiped with an optical wipe to remove impurities present on the surface. Since the glass samples are unpolished, the gap between the interfaces is large when stacked naturally, and a fixture is needed to reduce the gap between the samples as follows: stack the two pieces of glass up and down, and rub them slightly to remove the air at the interface until colored stripes at the interface can be observed, and the gap between the two pieces of glass is around $5 \mu\text{m}$ by calculating [27]. Then place the tightly fitting glass in the fixture, tighten the screws of the fixture to fix it, and ensure that the glass is leveled, as shown in Fig. 2(a). Weld strength testing with the MFM1200 Push-Pull Tester with a measuring range of 10–2000N and a probe displacement speed of $300 \mu\text{m/s}$. In the same operating conditions, 3 groups of repetitive experiments, take the average of the three results as the final strength test results to ensure the reliability of the data, the test method is shown in Fig. 2 (b).



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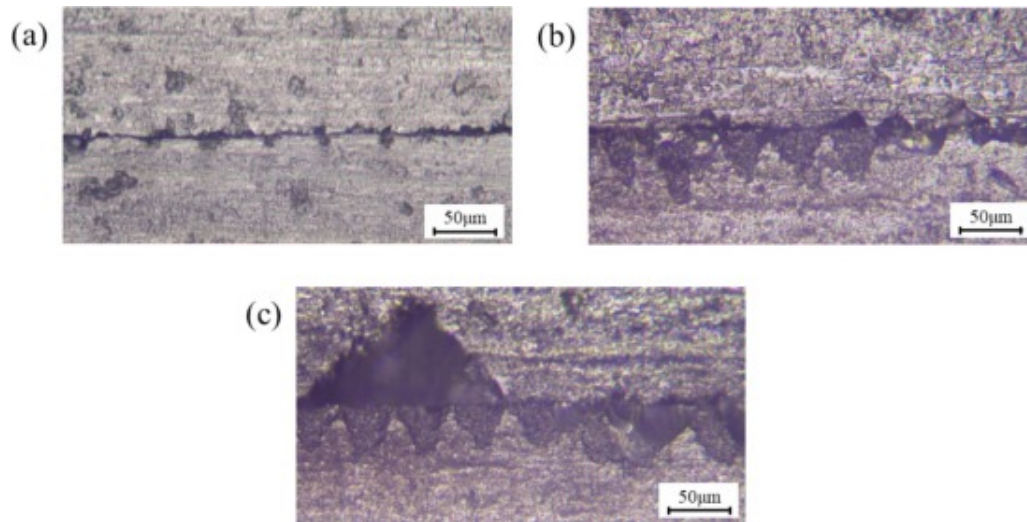
Fig. 2. Schematic diagrams of fixture(a) and strength test methods(b).

3. Results and discussion

3.1. Effect of laser power on weld cross-section morphology

Soda-lime glass has a low picosecond laser absorption of 1064nm, which causes nonlinear absorption only when the laser intensity is high enough, otherwise a reliable weld cannot be formed. Therefore, in order to determine the appropriate laser power, multi-pass scanning experiments were carried out with a scanning speed of 100mm/s, a repetition frequency of 50 kHz, and a focusing position of the upper surface of the bottom glass plate. The cross-sectional morphology of the welds with different powers is shown in Fig. 3. When the laser power is 1.5W, no continuous and uniform weld can be seen, and the black spots at the interface represent the local volume modification at the interface, at this time, the strength of the weld is extremely low, and the two pieces of glass can be separated by slight shaking, which represents that no effective connection is formed. When the laser power is 2.5W, the triangular modified area in the bottom glass can be clearly observed at the interface of the top and bottom glass, and a reliable connection has been formed at this time. At a laser power of 3.5 W, large-size defects appeared in

localized areas of the top glass plate, which could be attributed to the high stress generated at high power. Therefore, the laser power was set to 2.5 W in subsequent experiments in order to form a reliable connection and to minimize the generation of welding defects.



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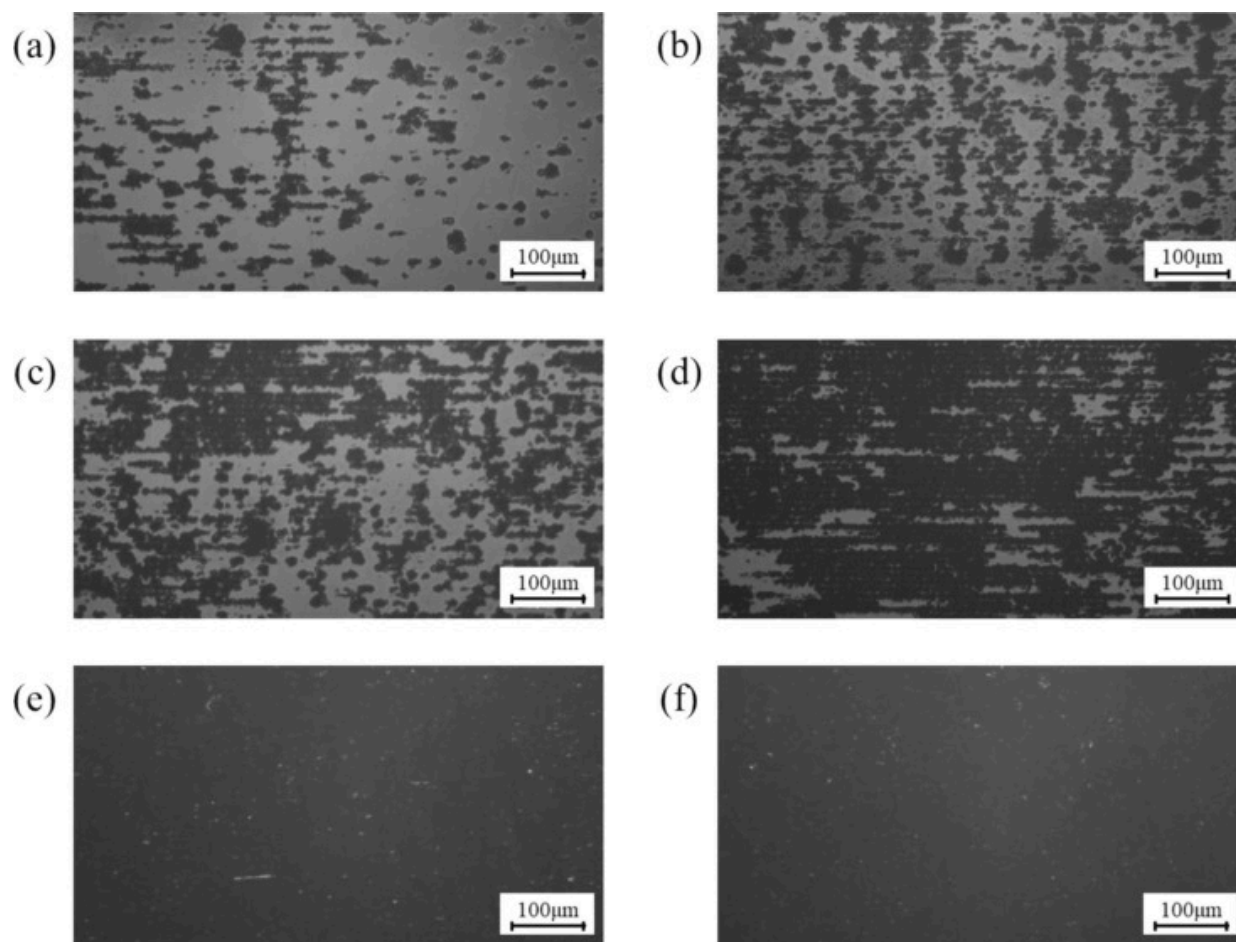
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Fig. 3. Cross-sectional morphology of welds with different power; (a) 1.5W, (b) 2.5W, (c) 3.5W.

3.2. Effects of the scanning times and scanning spacing on the welding strength

The actual size of the welded area is 10mm×5mm, the process needs to repeat the scanning of the welded area several times, the scanning times is set to 5~50, the scanning spacing of 10µm, different scanning times of the weld as shown in Fig. 4. It is observed that there are two different coloured areas, the gap between the top and bottom glass in the dark area has been effectively filled, while the light area represents that the gap has not been completely filled. When the number of scans is 5, most of the scanned area is light-coloured and the dark-coloured area is intermittent and very unevenly

distributed. As the scanning times increases, the proportion of dark-coloured areas gradually becomes larger, and when the number of scans is 20, the proportion of dark-coloured areas has reached 80%. When the number of scans is 25, basically the whole scanning area has been occupied by dark-coloured areas, but in the local area can still be observed in the light-coloured areas of scattered distribution, and as the number of scans continues to increase to 30, this situation has not been further improved.

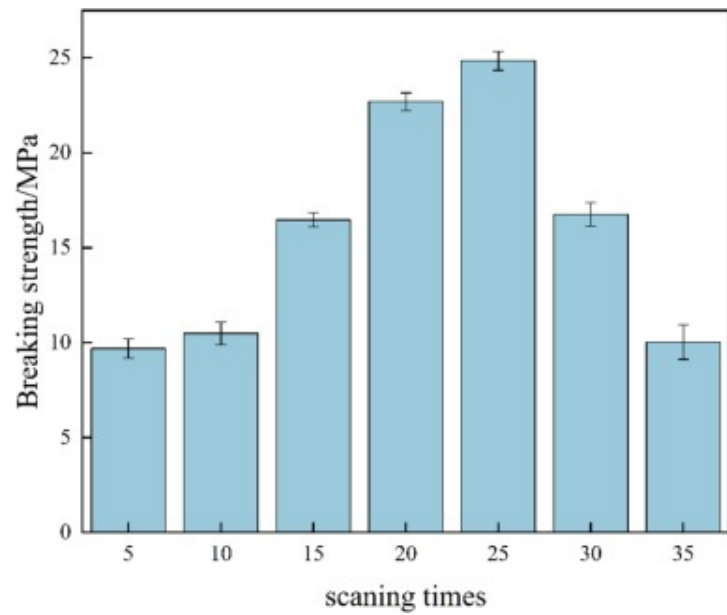


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Fig. 4. Surface of the weld obtained with different scanning times; (a) 5 times, (b) 10 times, (c) 15 times, (d) 20 times, (e) 25 times, (f) 30 times.

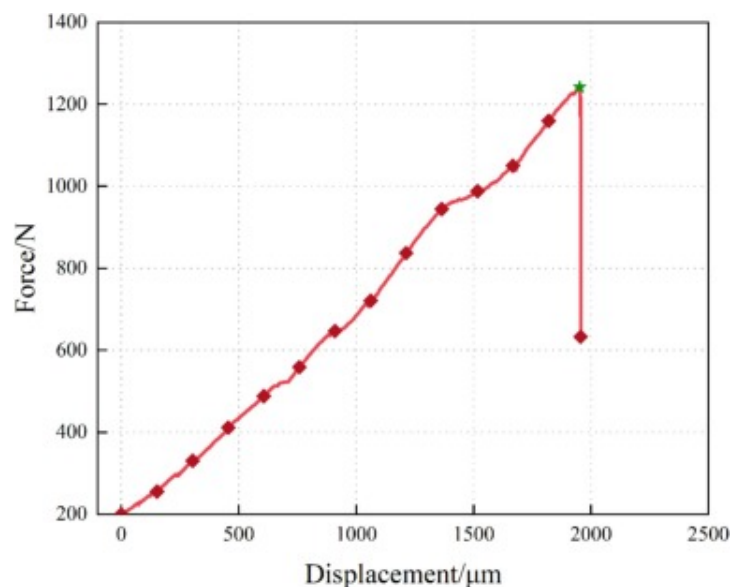
The breaking strength of the welds obtained with different scanning times was tested, as shown in Fig. 5. When the scanning times is 5, the breaking strength is only 9.69MPa, and with the increase of the scanning times, the breaking strength gradually increases. As shown in Fig. 6, at a scanning number of 25 times, the Breaking strength test results indicate a maximum breaking force of 1242.5N, a weld area of $10 \times 5 \text{ mm}^2$ and a corresponding weld breaking strength of 24.85MPa. If the scanning times continues to increase, the breaking strength decreases instead. It is worth noting that when the scanning times is further increased to 50, the two pieces of glass can be separated by a slight pull with the hand, meaning that an effective connection is not formed at this time. Picosecond laser welding of glass is mainly realized through the high energy density laser to melt the glass at the interface to fill the gap between the two, therefore, whether the gap between the top and bottom glass can be effectively filled is the main reason for the impact of the breaking strength. When the scanning times is small, the volume of glass melt is limited and not enough to completely fill the gap between the top and bottom glass, so the breaking strength is low. As the scanning times increases, the volume of the glass melt increases accordingly and the gap between the top and bottom glasses is gradually filled, so the breaking strength increases. However, when the scanning times is too high, due to the excessive expansion of the glass melt, a large stress is generated between the top and bottom glasses, which leads to a decrease in the strength of the weld and even prevents the formation of an effective connection.



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Fig. 5. Breaking strength under the different scanning times.



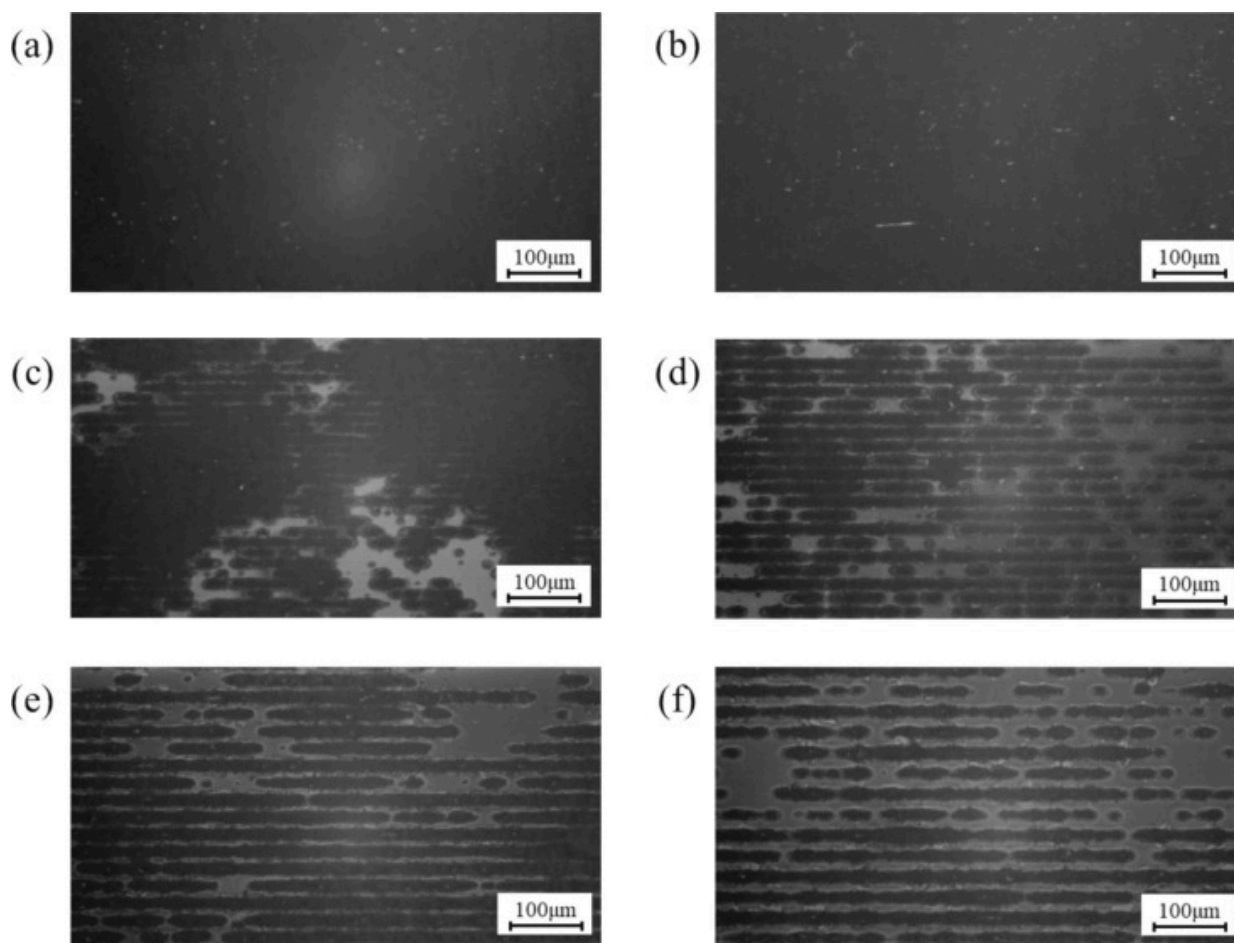
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Fig. 6. The test results of the weld strength with 25 scanning times.

Scanning spacing is also one of the main factors affecting the processing, the scanning spacing is too small means that the overlap between neighboring weld seams is large, the overlap area is processed several times, and this will also lead to a reduction in processing efficiency; on the contrary, the scanning spacing is too large means that the spacing between two neighboring weld seams is far away, and if there is not any overlap, it will lead to the failure of these areas to realize the connection. In order to obtain the appropriate processing parameters, the scanning spacing is set to 5~30μm, the scanning times is 25, and the photos of the weld surface obtained with different scanning spacing are shown in Fig. 7. When the scanning spacing is 5–10μm, the dark-colored areas are continuously and uniformly distributed at the entire scanning interface; when the scanning spacing is increased to 15μm, some light-colored areas appear, which means that these areas are not effectively filled, and the weld as a whole appears to be a

discontinuous feature; as the scanning spacing is further increased, the spacing between the two neighboring weld seams becomes more obvious.



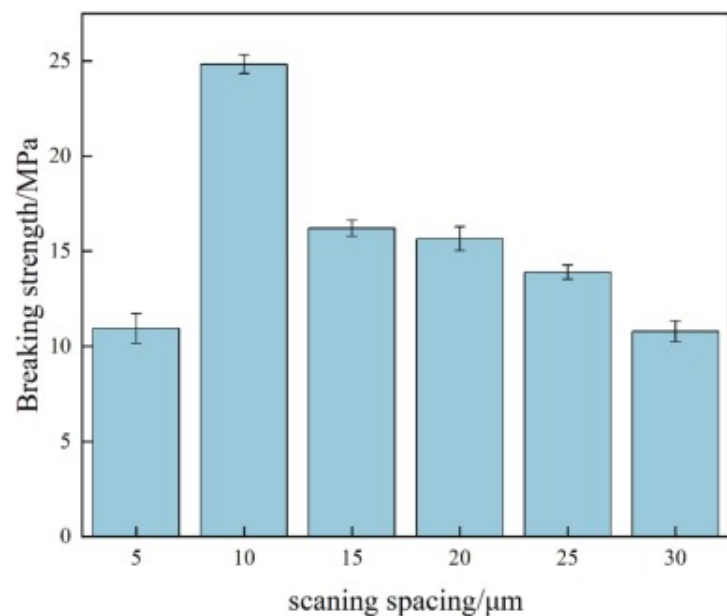
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Fig. 7. Surface of the weld obtained with different scanning spacing; (a) 5 μm , (b) 10 μm , (c) 15 μm , (d) 20 μm , (e) 25 μm , (f) 30 μm .

The breaking strength of the welds obtained at different scanning spacing are shown in [Fig. 8](#). It can be seen that the breaking strength of the scanning spacing of 10 μm is the

highest, 24.85 MPa, and then with the increase of the scanning spacing, the strength gradually decreases, and the strength decreases to 13.91 MPa at the scanning spacing of 25 μm , and the strength is only 10.80 MPa at the scanning spacing of 30 μm . The gap between the top and bottom glass in the whole scanning area is effectively filled when the scanning spacing is 10 μm , so the overall strength of the weld is more evenly distributed. The gap between the top and bottom glass is effectively filled in the whole scanning area with a scanning pitch of 10 μm , and the weld is uniformly distributed in the whole area, so higher strength is obtained. When the scanning spacing is 5 μm , although the entire gap is filled, the overlap between adjacent welds is large, and these overlapping parts are repeatedly scanned by the laser many times, resulting in an increase in stress in the local area, and thus the breaking strength of the weld is reduced. When the scanning spacing is 15 μm or even larger, the weld is not filled with the entire scanning area, and there is an un-lapped area between adjacent welds, and with the increase of scanning spacing, the un-lapped area gradually increases, and in fact, the effective area of the weld decreases, so the breaking strength is gradually reduced. Therefore, too large or too small scanning spacing is unfavorable to glass welding.



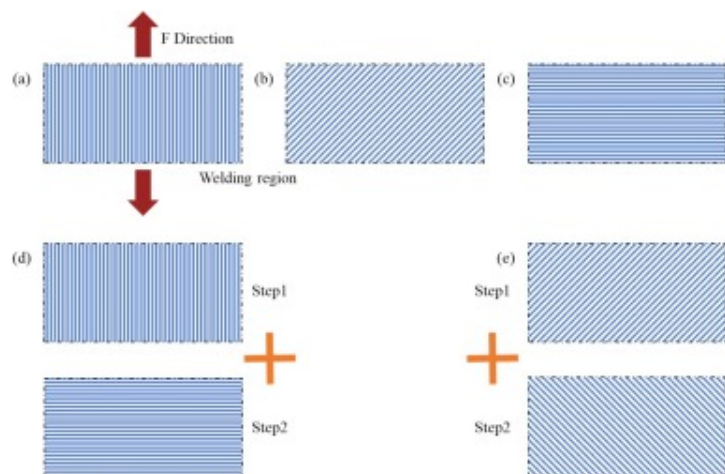
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Fig. 8. Breaking strength under the different scanning spacing.

3.3. Effects of scanning direction and focus position on the welding strength

In order to investigate the effect of laser scanning direction on weld strength, five different scanning modes were set up as shown in Fig. 9. Among them, the direction of the force is fixed in the breaking strength test, and the scanning angle is defined as the angle between the scanning direction and the direction of the force, which is 0° , 45° and 90° as shown in Fig. 9(a)(b)(c), respectively, and the scanning direction is fixed and 25 repetitions of the scanning are performed during the processing of the above three scanning methods. The remaining two are cross scanning modes, Fig. 9(d) shows 0° and 90° cross scanning, in which 0° scanning and 90° scanning are carried out alternately, and Fig. 9(e) shows 45° and 135° cross scanning, in which 45° scanning and 90° scanning are also carried out alternately.

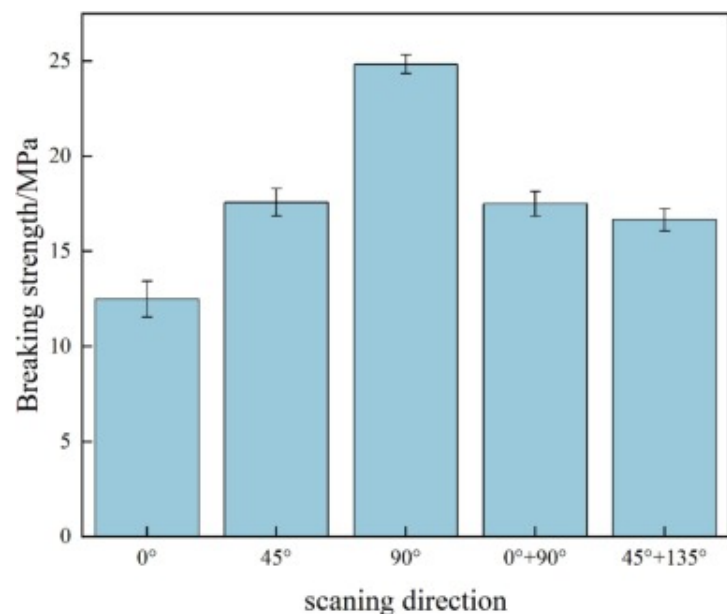


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Fig. 9. Schematic diagram of the scanning direction; (a) 0° , (b) 45° , (c) 90° , (d) $0^\circ+90^\circ$, (e) $45^\circ+135^\circ$.

The breaking strengths under different scanning direction are shown in Fig. 10. As the scanning direction goes from 0° to 45° and then to 90° , the corresponding weld strength increases gradually, while the strength of the weld obtained by the cross-scanning method of $0^\circ+90^\circ$ is about the average value of the unidirectional scanning methods of 0° and 90° , and the strength of the weld obtained by the cross-scanning method of $45^\circ+135^\circ$ is basically the same as that obtained by the unidirectional scanning method of 45° . Therefore, for glass welding, the scanning direction perpendicular to the stress direction of the weld is more favorable to improve the overall strength of the weld.



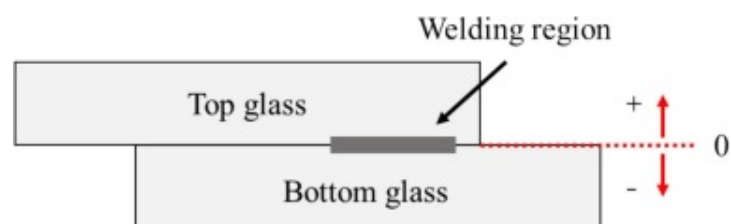
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Fig. 10. Breaking strength under the different scanning direction.

The focus position is also one of the key factors affecting the quality of the weld. The schematic of out-of-focus amount is shown in Fig. 11. The upper glass plate is closely fit to the lower glass plate, and the upper surface of the lower glass plate is located at the position of 0, which is a positive value above this plane, and on the contrary, it is a negative value below this plane. In this experiment, the welds obtained from five different focal positions of +0.2 mm, +0.1 mm, 0, -0.1 mm, and -0.2 mm were used for strength testing as shown in Fig. 12. The breaking strength of the weld is highest when the amount of defocus is 0. Focus position moving downward or downward reduces the strength of the weld, and the trend is similar. Glass welding is mainly realized by the irreversible expansion generated during the local material melting and re-solidification process near the focus, and the focus position is prone to produce more melt when it is

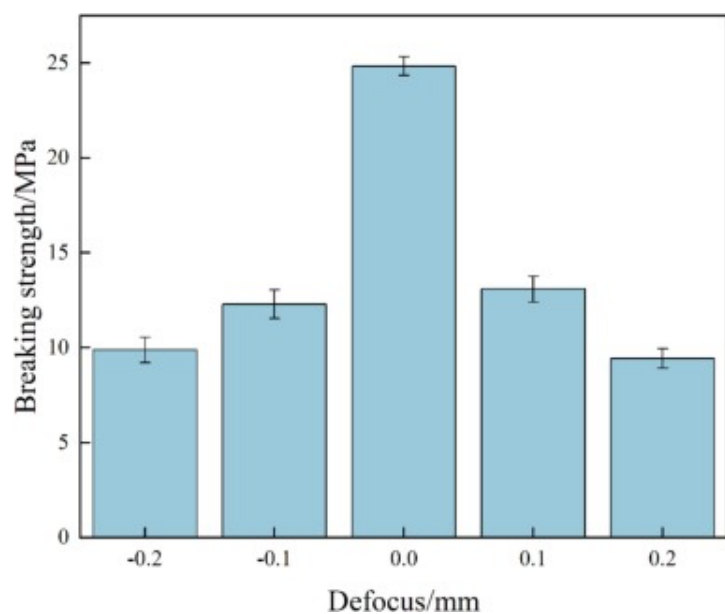
located at the interface of the two glass plates, thus effectively filling the gap between the glass plates.



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Fig. 11. Schematic of the defocus.



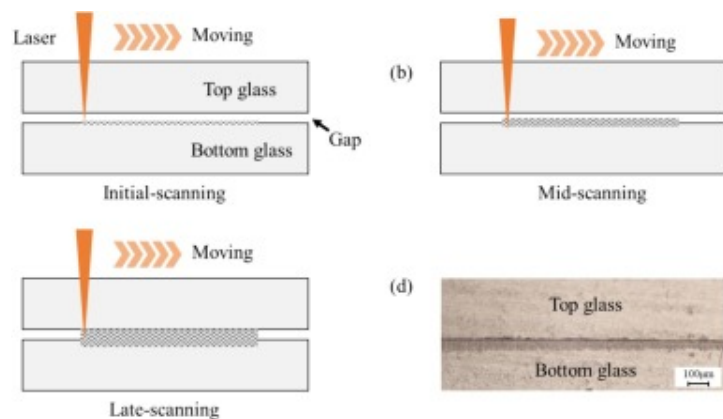
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Fig. 12. Breaking strength under the different defocus.

3.4. Analysis of glass welding mechanism

In the process of multiple scan welding, there are three main stages: the initial-scanning stage, the mid-scanning stage and the late-scanning stage, as shown in Fig. 13. In the initial-scanning stage, the focus acts on the upper surface of the bottom glass, causing the glass near the upper to be modified or melted and then solidified, which makes it easier to absorb the subsequent laser energy. During the initial-scanning stage, the melt expands to a limited volume, which is not sufficient to completely fill the gaps between the glass plates, thus requiring more scans. Under the continued action of the subsequent laser energy, a larger volume of material undergoes melting and re-solidification, which further reduces the gap until it is completely filled, forming a reliable weld, as shown in Fig. 13. During welding, there is a large amount of stress distributed in and around the glass weld, and an increase in the scanning times will inevitably introduce additional stress, and the accumulated stress leads to defects such as cracks in the sample, or even weld failure. Therefore, the appropriate number of scans is critical.



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Fig. 13. Schematic diagram of welding process. (a) initial-scanning, (b) mid-scanning, (c) late-scanning, (d) weld cross-section morphology.

4. Conclusion

Welding of non-optical contact soda-lime glass was realized by multiple scans using a 12ps laser. A breaking strength of 24.85MPa was obtained by process optimization in a weld area of 10mm×5mm. The main conclusions are as follows:

- (1) Appropriate laser power is the basis for forming a reliable connection. Low power will lead to insufficient energy to form a weld; high power will cause an increase in defects, and the optimal power is 2.5W.
- (2) The scanning times and scanning spacing have a large impact on the strength of the weld. The scanning times on the one hand affects the melt volume, thus determining whether the gap between the glass can be completely filled; on the other hand, affects the size of the stress generated during the welding process, too few or too many scanning times will lead to a reduction in strength. Scanning spacing mainly affects the overlap rate between neighboring welds, too large an overlap rate means that the overlap part is repeatedly ablated, causing an increase in stress, and too low an overlap rate may lead to the gap can not be completely filled, the overall effective area of the weld is reduced. The optimum scanning times and scanning spacing are 25 times and 10μm, respectively.
- (3) In addition, the scanning direction and focus position also affect the weld strength. The breaking strength is maximum when the scanning direction is perpendicular to the force direction. The focus position is within 0.2mm above and below the glass contact interface to form an effective weld. The strength is 24.85MPa when the focus position is in interface.

CRedit authorship contribution statement

Yichen Huang: Writing – review & editing. **Zihao Zhang:** Writing – original draft. **Xu Wang:** Methodology. **Liqun Li:** Supervision. **Lijun Yang:** Supervision. **Mingyu Li:** Funding acquisition. **Wenbo Zhu:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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